NaturArv3D Data Pipeline

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Introduction to `ome_zarr` CLI concept

- Have made PRs to `ome-zarr-py` repo, $2/4$ merged
  - TIFF stack export command: `ome-zarr-py#678`
  - Built-in support for `imagecodecs` codecs: `ome-zarr-py#679`
  - Even if not merged: shows how easily conversion tool can be made and maintained
- Reminder, importantly: many codecs are still not *standardised* in Zarr 3
  - No standardised lossy codecs as of 2026-10-06
  - Current proposal relies on one specific codec implementation (`imagecodecs`)
  - Upgrade path to standardised codec is expected, but not guaranteed
  - Proposed JPEG XL Extension Specification: `zarr-extensions#70`

Alternatives remain:

- DICOM with JPEG-LS or JPEG 2000 compression
- Uncompressed (perhaps losslessly compressed) TIFF stack

Demo `ome_zarr` CLI

1. Install `uv`:
   - *Some Python people may already have it installed*
     - *≈ 30% market share in October 2025: caveats apply, see source.*
     - *90.4K stars on GitHub (2026-10-05)*
   - Windows: `winget install --id=astral-sh.uv  -e`
   - Linux/MacOS: `curl -LsSf https://astral.sh/uv/install.sh | sh`
2. Install `ome-zarr-py` CLI:
   - `uv tool install --with imagecodecs-zarr git+https://github.com/JeppeKlitgaard/ome-zarr-py@feat-cli-convert`
3. Stream OME-Zarr and convert to TIFF stack (here we pick second level):
   - `ome_zarr export --format tiff-stack --level 2 --overwrite LINK_OR_LOC_OF_OME_ZARR output_tiffs`
   - Example:
     - `ome_zarr export --format tiff-stack --level 2 --overwrite "https://s3.g.s4.mega.io/aidjqnzhivz2k6kyxukiymjgfb4wxowhengdp/naturarv3d/CIL_basic_binned_2x2/NaturArv3D_format_viability_study/OME-Zarr/Canislupus_CZM_712__jpegxl_l_0_5__scales_2_4_8_16.ome.zarr" output_tiffs`
4. *Optional: Open in Napari (`uvx --with pyqt6 napari output_tiffs`)*

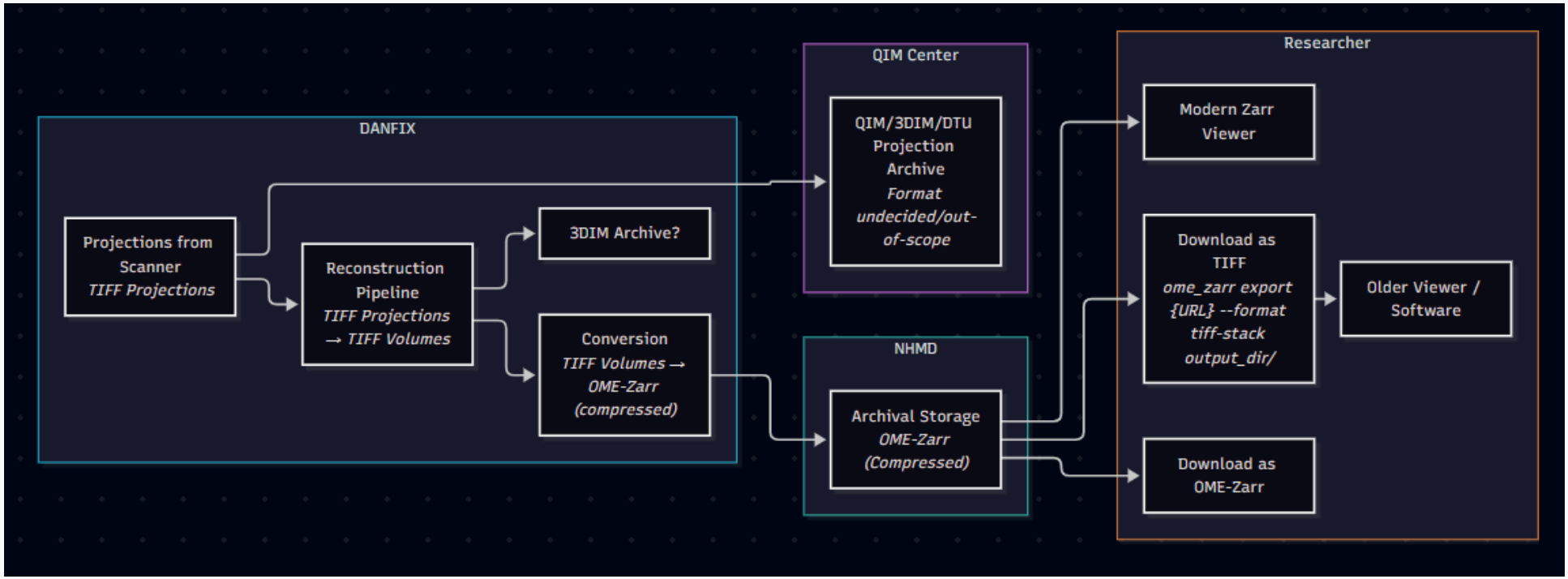
Tool Demo

Notes on tool

- Nice:
  - Streams in compressed format, so network friendly
  - Streams, so memory friendly
  - Emits standard TIFF stack
  - Preserves voxel size metadata
- Just a prototype
  - Intended to prove feasibility
  - Could just as well be a little graphical interface, or both

Data Flow Chart

- Very rough flow chart of data pipeline
- Mostly just included for later reference
  - *Exact timing of conversions and where compression happens is still TBD*.
  - Right-most half may be instructive

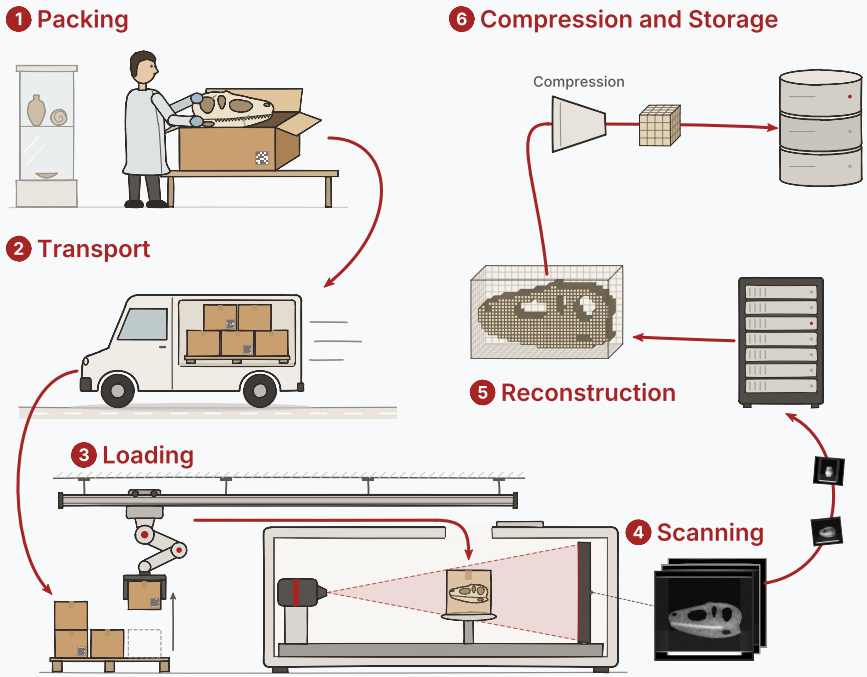


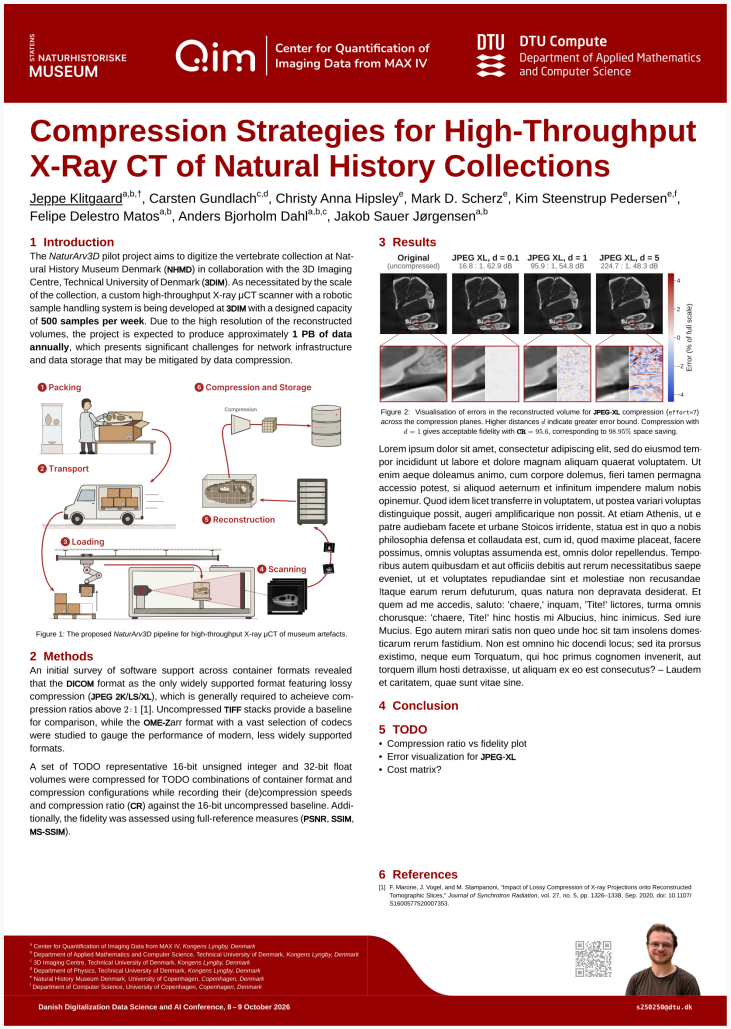
Summary

- Initially thought mostly about codecs
- Container format turned out to be more important
- May need to revisit codec when we get closer to implementation stage
- Don’t have to make decision now?

Aside: Poster at D3A 4.0 Conference (October 8 – 9)

- Bit last minute, *apologies*
- Hopefully will be engaging to people
- Non-finished version to the right
  - *(due to monotonicity of time)*





NATURHISTORISKE MUSEUM

Center for Quantification of Imaging Data from MAX IV

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Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

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1 Introduction

The *NaturArv3D* pilot project aims to digitize the vertebrate collection at Natural History Museum Denmark (**NHMD**) in collaboration with the 3D Imaging Centre, Technical University of Denmark (**3DIM**). As necessitated by the scale of the collection, a custom high-throughput X-ray µCT scanner with a robotic sample handling system is being developed at **3DIM** with a designed capacity of **500 samples per week**. Due to the high resolution of the reconstructed volumes, the project is expected to produce approximately **1 PB of data annually**, which presents significant challenges for network infrastructure and data storage that may be mitigated by data compression.

Figure 1: The proposed *NaturArv3D* pipeline for high-throughput X-ray µCT of museum artefacts.

2 Methods

An initial survey of software support across container formats revealed that the **DICOM** format as the only widely supported format featuring lossy compression (**JPEG 2K/LS/XL**), which is generally required to acheieve compression ratios above $2:1$ [1]. Uncompressed **TIFF** stacks provide a baseline for comparison, while the **OME-Z**arr format with a vast selection of codecs were studied to gauge the performance of modern, less widely supported formats.

A set of TODO representative 16-bit unsigned integer and 32-bit float volumes were compressed for TODO combinations of container format and compression configurations while recording their (de)compression speeds and compression ratio (**CR**) against the 16-bit uncompressed baseline. Additionally, the fidelity was assessed using full-reference measures (**PSNR**, **SSIM**, **MS-SSIM**).

3 Results

Original (uncompressed) | JPEG XL, d = 0.1 16.8 : 1, 62.9 dB | JPEG XL, d = 1 95.9 : 1, 54.8 dB | JPEG XL, d = 5 224.7 : 1, 48.3 dB

Figure 2: Visualisation of errors in the reconstructed volume for **JPEG-XL** compression (`effort=7`) across the compression planes. Higher distances d indicate greater error bound. Compression with $d = 1$ gives acceptable fidelity with **CR** ≈ 95.6, corresponding to 98.95% space saving.

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4 Conclusion

5 TODO

- Compression ratio vs fidelity plot
- Error visualization for **JPEG-XL**
- Cost matrix?

6 References

[1] F. Mennne, J. Vogel, and M. Stampanoni, "Impact of Lossy Compression of X-ray Projections onto Reconstructed Tomographic Slices," *Journal of Synchrotron Radiation*, vol. 27, no. 5, pp. 1326–1338, Sep. 2020, doi: 10.1107/S1600577520007353.

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Danish Digitalization Data Science and AI Conference, 8 – 9 October 2026

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Closing Remarks

- *Everything (?)* available on private GitHub repo under `qim-center` org
  - https://github.com/qim-center/consult-xct-compression
    - QIM + Close friends already have access
    - Others can ask for inclusion
  - Slides and source code for every meeting presentation
  - Source code and outputs for all experiments and demos
    - Note: Obviously no volumetric data included
  - Poster including source code
  - Unfinished report that has been abandoned